UNIVERSITY OF BOHOL

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A documentation for

CPEP 311A

Midterm Project

"Infrared-Controlled Automatic Door Prototype with Transistor Logic and Servo Control"

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Course and Section:

BSCpE – 3

1. INTRODUCTION TO THE PROJECT

Malls, schools, offices and stores use automatic doors so people can enter without touching the door, especially when their hands are full. A manual door needs someone to push or pull it, and it is often left open or closed at the wrong time. An automatic door removes these problems by detecting a person, opening by itself, and closing again after the person has passed.

This project is an Automatic Door built entirely from discrete electronic components, without a microcontroller. An FC-51 infrared (IR) obstacle sensor detects an approaching person. A transistor delay circuit, made of a 1000 µF capacitor and a 30 kΩ resistor chain, keeps the door open for about five seconds after the person leaves the sensor so that the door does not close on the person. Transistor logic (Q1, Q2 and Q3) drives an LTP-587G 16-segment display that shows a check mark (✓) when the door is open and an X when the door is closed. An NE556 dual timer generates a 50 Hz clock and a pulse-width-modulated (PWM) signal that moves an SG90 servo motor, which acts as the door panel between 0° (closed) and 180° (open).

The project applies the topics covered in the midterm period: transistor switching, logic inversion, truth tables, Boolean expressions and their simplification, RC timing, and 555/556 timer circuits. The prototype was assembled on a breadboard and mounted on a scale model of a door entrance. timing, and

1.1 Objectives

- To design and build an automatic door that opens when a person is detected and closes after a delay.
- To show the door status clearly using a 16-segment display (✓ for open, X for closed).
- To apply truth tables, Boolean expressions and Boolean simplification to the control logic of the circuit.
- To control an SG90 servo motor with an NE556 timer and calibrate its open (180°) and closed (0°) positions using potentiometers.

1.2 Scope and Limitations

The prototype is a scale model powered by a 5 V, 2 A supply. It detects an object using one FC-51 IR sensor, so it responds to any object within the sensor range and cannot tell a person from other objects. Strong sunlight can affect the IR sensor. The five-second delay depends on the tolerance of the capacitor and resistors and may vary slightly between builds.

2. PROJECT DESCRIPTION

The Automatic Door has four working parts: the sensing and delay section, the logic and display section, the timing and PWM section, and the servo door panel. The operating sequence is as follows.

1. Idle (door closed). No object is in front of the sensor. The delay capacitor C2 is fully charged, so Q4 conducts and the Master Signal (M) is HIGH. Q1 turns on and lights the X. Q2 turns on, which turns Q3 off, so the check mark is dark. Pot 2 sets the servo to 0° (closed).

2. Person detected (door opens). The IR output goes LOW. Diode D1 discharges C2 immediately, so Q4 turns off and M goes LOW. Q1 and Q2 turn off, the X goes dark, and the collector of Q2 rises to 5 V through its 10 kΩ pull-up. This turns Q3 on and lights the check mark. With Q3 on, the Pot 1 branch controls the PWM signal and the servo moves to 180° (open).

3. Person leaves (delay). The IR output returns HIGH and D1 is reverse-biased, so C2 starts charging through the 30 kΩ resistor chain. The door stays open for about five seconds.

4. Door closes. When C2 has charged enough to turn Q4 on again, M goes HIGH. The display changes back to X, Q3 turns off, the 1 kΩ booster resistor and diode D2 feed 5 V into the Pot 2 branch, and the servo returns to 0° (closed).

2.1 Main Functional Blocks

Block	Function
Power and filtering	5 V / 2 A supply with a 1000 µF capacitor across the rails to smooth the supply and absorb the servo current spikes.
FC-51 IR sensor	Detects an object in front of the gate. Its OUT pin goes LOW when an object is detected.
Q4 delay buffer	Emitter follower with a 1000 µF capacitor, 1N4148 diode and 30 kΩ resistor chain. It gives an instant reaction to a vehicle and a delay of about 5 s before closing.
Q1, Q2, Q3 logic	Q1 drives the X segments. Q2 inverts the Master Signal. Q3, driven by Q2, drives the check-mark segments and boosts the servo timing network.
LTP-587G display	Common-anode 16-segment display connected to 5 V. Shows X (closed) or ✓ (open).

Block	Function
NE556 Timer 1	Astable oscillator of about 50 Hz (1.44 / ((100 kΩ + 2 × 100 kΩ) × 0.1 µF) ≈ 48 Hz) that triggers Timer 2.
NE556 Timer 2	Generates the PWM pulse that sets the servo angle. The pulse width is about 1.1 × R × C, where R is set by Pot 1 or Pot 2.
SG90 servo	Barrier arm. Red wire to 5 V, brown to GND, orange (signal) to Pin 9.

3. DIAGRAMS

3.1 System Block Diagram

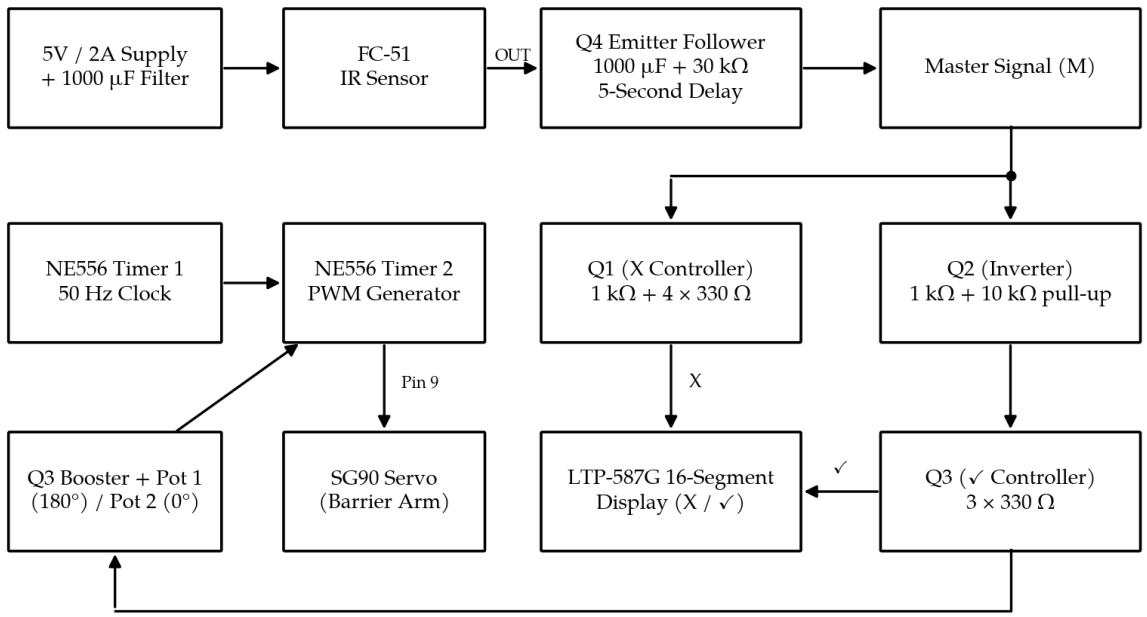


Figure 1. System block diagram of the Carpark Toll Gate

3.2 Circuit Diagrams

The complete circuit is divided into three parts so that each can be read clearly. All three parts share the same +5 V and GND rails.

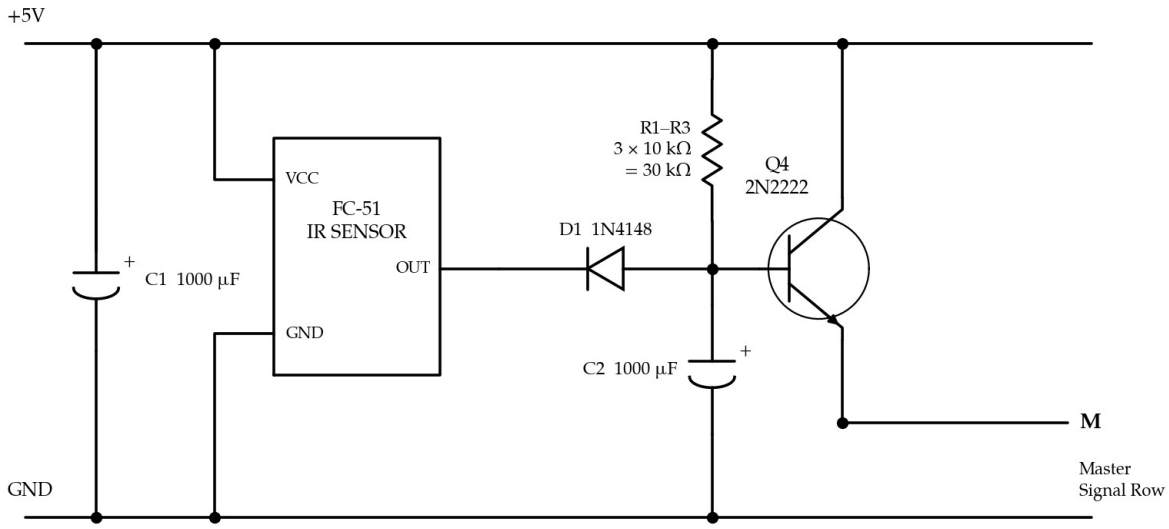


Figure A. Power filter, FC-51 IR sensor and 5-second delay buffer

Figure 2. Power filter, FC-51 IR sensor and 5-second delay buffer (Q4)

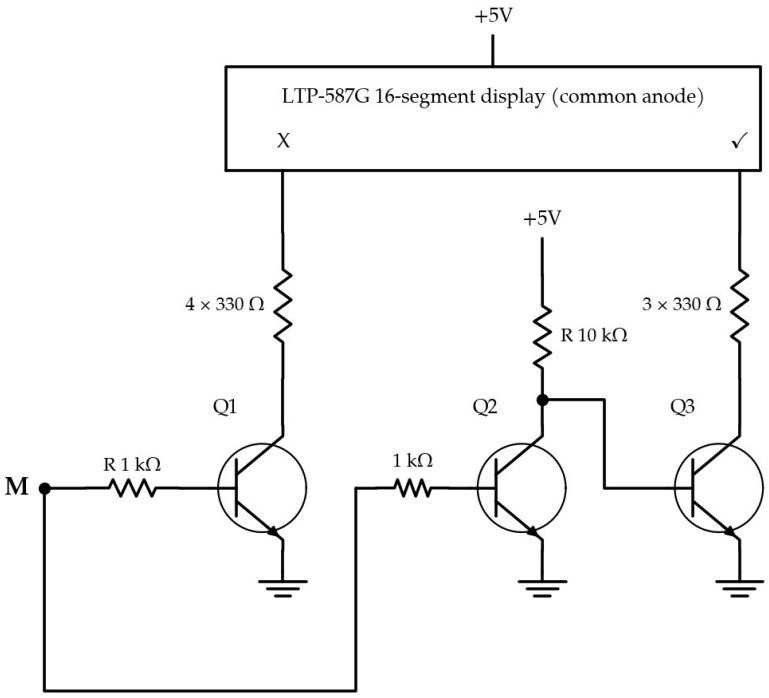


Figure B. Display controllers Q1 and Q3 with logic inverter Q2

Figure 3. Display controllers Q1 and Q3 with logic inverter Q2

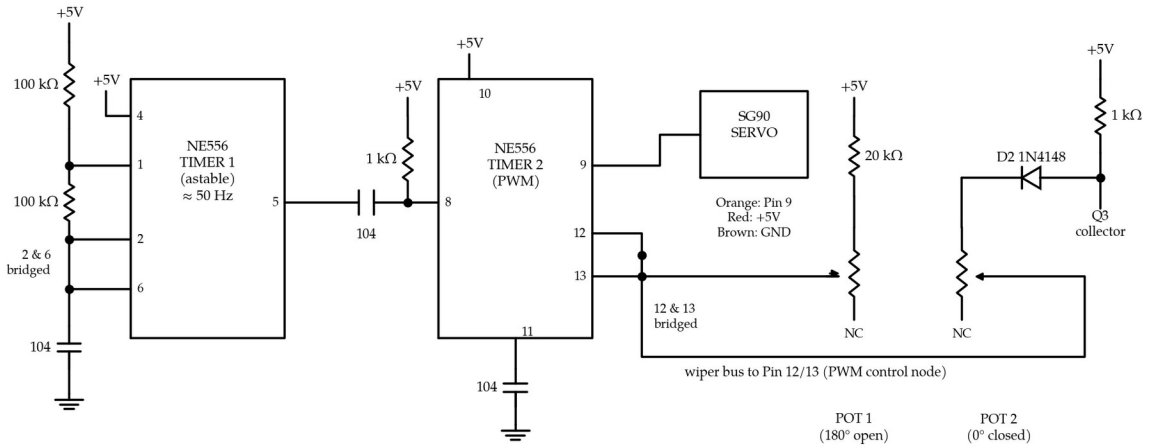


Figure C. NE556 timers (50 Hz clock and PWM generator), calibration potentiometers and SG90 servo

Figure 4. NE556 timers, calibration potentiometers and SG90 servo

Note: The 1000 µF capacitor C1 across the power rails, the 104 decoupling capacitors on Pins 14 and 11 of the NE556, and the connections of Pins 14, 4 and 10 to 5 V and Pin 7 to GND are described in Section 4 and are not repeated in the figures.

4. DOCUMENTATION: CIRCUIT CONSTRUCTION PROCEDURE

The following is the build flow that was followed to assemble and calibrate the prototype.

4.1 Power Setup and Filtering

1. Connect the 5 V / 2 A power supply to the main positive (+) and negative (–) power rails of the breadboard.
2. Insert a 1000 µF electrolytic capacitor across the main power rails. The long leg (+) goes to 5 V and the short leg (marked with the minus stripe) goes to GND.

4.2 IR Sensor and 5-Second Delay Buffer

1. Connect the FC-51 IR sensor VCC to 5 V and GND to GND. Plug the OUT pin into an empty row (the IR Output Row).
2. Place Q4 (2N2222) with the flat side facing you. Connect the Collector (right leg) to 5 V.
3. Connect the Emitter of Q4 (left leg) to a new empty row. This is the Master Signal Row (M).
4. Connect a 1000 µF capacitor with its long leg (+) to the Base (middle leg) of Q4 and its short leg (–) to GND.
5. Connect a 1N4148 diode with its anode (no stripe) to the Base of Q4 and its cathode (stripe) to the IR Output Row.
6. Chain three 10 kΩ resistors in series to make a 30 kΩ path from the 5 V rail to the Base of Q4.

4.3 16-Segment Display and Logic Inverter

1. Connect the common-anode pins of the LTP-587G display directly to 5 V.
2. Q1 (X controller): Emitter to GND. Base to the Master Signal Row through a 1 kΩ resistor. Collector to the four X segment pins through four separate 330 Ω resistors.
3. Q2 (inverter): Emitter to GND. Base to the Master Signal Row through a 1 kΩ resistor. Collector to 5 V through a 10 kΩ pull-up resistor.
4. Q3 (✓ controller): Emitter to GND. Base directly to the Collector of Q2 with a jumper wire. Collector to the three check-mark segment pins through three separate 330 Ω resistors.

4.4 Timer 1 (50 Hz Heartbeat Clock)

1. Insert the NE556 IC. Connect Pins 14, 4 and 10 to 5 V and Pin 7 to GND.
2. Add noise filters: a 104 ceramic capacitor across Pin 14 and Pin 7, and another 104 capacitor from Pin 11 to GND.
3. Bridge Pin 2 and Pin 6 with a short jumper wire, then connect a 104 capacitor from Pin 2 to GND.
4. Connect a 100 kΩ resistor from 5 V to Pin 1, and a second 100 kΩ resistor from Pin 1 to Pin 2.
5. Connect a 1 kΩ resistor from 5 V to Pin 8 to hold the trigger input HIGH.
6. Connect Pin 5 to Pin 8 through a 104 ceramic capacitor.

4.5 Timer 2 (PWM) and Q3 Servo Booster

1. Bridge Pin 12 and Pin 13 with a short jumper wire and connect a 104 capacitor from Pin 12/13 to GND.
2. Q3 voltage booster: connect a 1 kΩ resistor from 5 V to the Collector of Q3. Connect the anode of a 1N4148 diode to that Collector and its cathode to a new empty row.
3. Pot 2 (0° closed): Left leg to the diode cathode row, Middle leg to Pin 12/13, Right leg empty.
4. Pot 1 (180° open): connect a 20 kΩ resistor (two 10 kΩ in series) from 5 V to the Left leg, Middle leg to Pin 12/13, Right leg empty.
5. Servo motor: SG90 red wire to 5 V, brown wire to GND, orange wire to Pin 9.

4.6 System Calibration

1. **Calibrate 180° (open):** Place an object in front of the IR sensor. The 1000 µF capacitor drains instantly, the display shows ✓ and the servo receives the signal from Pot 1. Turn Pot 1 until the servo horn reaches the desired 180° open position.
2. **Calibrate 0° (closed):** Remove the object and wait about 5 seconds for the 30 kΩ chain to charge the capacitor. The display switches to X, Q3 turns off and 5 V passes through the booster into Pot 2. Turn Pot 2 until the servo returns to the exact 0° closed position.

5. TRUTH TABLE

The control logic has two inputs and one internal output (the Master Signal):

- **A** = IR sensor condition. A = 1 when an object (vehicle) is detected and A = 0 when no object is present. (The FC-51 pin itself goes LOW on detection; A is defined as the detection condition.)

- **D** = delay capacitor state at Q4. D = 1 when the 1000 µF capacitor is charged (no object for at least about 5 seconds) and D = 0 when it is discharged or still charging.
- **M** = Master Signal (emitter of Q4). M = 1 when HIGH.
- **Q1, Q2, Q3** = 1 when the transistor is ON (conducting). **X** and ✓ = 1 when that part of the display is lit.

Because A = 1 forces the diode D1 to discharge the capacitor immediately, the case A = 1 and D = 1 can never happen in the real circuit. It is marked d (don't care).

Table 1. Truth table of the Carpark Toll Gate

A	D	M	Q1	Q2	Q3	X	✓
0	0	0	0	0	1	0	1
0	1	1	1	1	0	1	0
1	0	0	0	0	1	0	1
1	1	d	d	d	d	d	d

Table 2. System status for each input condition

A	D	Display	Servo / Gate	Meaning
0	0	✓	180° (open)	No object, capacitor still charging. This is the 5-second delay after a vehicle leaves, and also the start-up condition.
0	1	X	0° (closed)	Idle state. No vehicle and the delay has finished.
1	0	✓	180° (open)	Vehicle detected. The capacitor is drained instantly and the gate opens.
1	1	d	d	Not physically possible.

6. BOOLEAN EXPRESSION AND SOLUTION

6.1 Derivation of the Expressions

Q4 is an emitter follower. It conducts only when there is no object (A′) and the capacitor has charged (D). Therefore the Master Signal is:

$$M = A' \cdot D$$

Q1 and Q2 receive M through 1 kΩ resistors, so both are ON when M = 1. The X segments are lit when Q1 is ON:

$$Q1 = M \qquad Q2 = M \qquad X = Q1 = M = A' \cdot D$$

Q2 is a common-emitter inverter. Its collector is pulled up to 5 V by the 10 kΩ resistor, so the collector is HIGH when Q2 is OFF and LOW when Q2 is ON:

$$Q2\ \text{collector} = M'$$

Q3 gets its base directly from the collector of Q2, so Q3 is ON when M is 0. The check mark is lit when Q3 is ON:

$$Q3 = M' \qquad ✓ = Q3 = M'$$

6.2 Solution (Substitution and De Morgan's Theorem)

Substituting M = A′ · D into the expression for the check mark and applying De Morgan's theorem:

$$✓ = M' = (A' \cdot D)' = (A')' + D' = A + D'$$

The servo is at 180° (open) whenever Q3 is ON, so Gate Open = ✓ = A + D′. The solved expressions are checked against the truth table below.

Table 3. Verification of the solved expressions

A	D	M = A′ · D	X = M	✓ = A + D′
0	0	1 · 0 = 0	0	0 + 1 = 1
0	1	1 · 1 = 1	1	0 + 0 = 0
1	0	0 · 0 = 0	0	1 + 1 = 1
1	1	d	d	d

Every row agrees with Table 1, so the derived expressions describe the circuit correctly.

7. BOOLEAN SIMPLIFICATION

7.1 Karnaugh Map

The Karnaugh map for the Master Signal M is shown below. The cell A = 1, D = 1 is marked d because it cannot occur.

Table 4. Karnaugh map for M

A \ D	0	1
0	0	1
1	0	d

The 1 in the cell A′D can be grouped with the don’t-care cell AD to form a group of two. This group is covered by D alone (A drops out because it takes both values inside the group).

$$M = D$$

7.2 Algebraic Simplification

The same result is obtained with Boolean algebra. Since the term A · D can never occur (A · D = 0), it may be added to the expression without changing it:

$$M = A' \cdot D$$

$$M = A' \cdot D + A \cdot D \qquad \text{(adding the impossible term } A \cdot D = 0\text{)}$$

$$M = D \cdot (A' + A) \qquad \text{(factoring out D)}$$

$$M = D \cdot 1 \qquad (A' + A = 1)$$

$$M = D$$

For the check mark, using M′ = D′:

$$✓ = M' = D'$$

The unsimplified form A + D′ gives the same result, because when A = 1 the capacitor is always discharged (D = 0, so D′ = 1) and A + D′ = D′ by the absorption law.

7.3 Final Simplified Expressions

Output	Original	Simplified
Master Signal (M)	A′ · D	D
X (gate closed)	A′ · D	D
✓ (gate open)	A + D′	D′

This shows that the whole circuit depends only on the state of the delay capacitor: the gate is closed (X) when the capacitor is charged, and it is open (✓) when the capacitor is discharged or

still charging. Because the IR sensor discharges the capacitor instantly, the sensor effectively controls D, and the five-second delay is the time D needs to return to 1.

7.4 Timing Calculations

- **Delay:** The time constant of the delay network is $\tau = R \times C = 30\ k\Omega \times 1000\ \mu F = 30$ s. Q4 and the display transistors switch when the capacitor reaches only a small fraction of the supply voltage, which gives a delay of roughly 5 seconds in the prototype. The exact value depends on component tolerance.
- **Timer 1 frequency:** $f = 1.44 / ((R1 + 2 \times R2) \times C) = 1.44 / ((100\ k\Omega + 2 \times 100\ k\Omega) \times 0.1\ \mu F) \approx 48$ Hz, which is close to the 50 Hz needed for servo control (period ≈ 20 ms).
- **Timer 2 pulse width:** $t = 1.1 \times R \times C$ with C = 0.1 µF. A pulse of about 1 ms ($R \approx 9.1\ k\Omega$) corresponds to the closed position and about 2 ms ($R \approx 18\ k\Omega$) to the open position of an SG90 servo. The potentiometers are adjusted during calibration until the horn reaches the exact positions.

8. MATERIALS

Table 5. List of materials

Item	Specification	Qty	Purpose
Breadboard	830 tie-points	1	Circuit assembly
Power supply	5 V / 2 A adapter	1	Main power
IR sensor module	FC-51 obstacle sensor	1	Vehicle detection
Transistor	2N2222 (NPN)	4	Q1, Q2, Q3, Q4
Diode	1N4148	2	D1 (delay), D2 (booster)
Electrolytic capacitor	1000 µF, 16 V	2	Power filter, 5 s delay
Ceramic capacitor	104 (0.1 µF)	5	Timer timing and noise filtering
16-segment display	LTP-587G (common anode)	1	Gate status (X / ✓)
Timer IC	NE556 dual timer	1	50 Hz clock and PWM
Servo motor	SG90 micro servo	1	Barrier arm

Item	Specification	Qty	Purpose
Trimmer potentiometer	10 kΩ (marked 103)	2	Calibration of 0° and 180°
Resistor	10 kΩ	6	30 kΩ chain, Q2 pull-up, 20 kΩ for Pot 1
Resistor	1 kΩ	4	Q1 and Q2 bases, Pin 8 pull-up, Q3 booster
Resistor	330 Ω	7	Display segment current limiting
Resistor	100 kΩ	2	Timer 1 frequency setting
Jumper / hook-up wires	Assorted, with Dupont servo header	1 set	Connections
Model materials	Illustration board, white paper, black tape	1 set	Carpark entrance model

9. COST

The prices below are estimated local prices in Philippine pesos (PHP). They should be replaced with the actual amounts on the receipts of the group before submission.

Table 6. Cost of materials

Item	Qty	Unit Price (PHP)	Amount (PHP)
Breadboard (830 points)	1	100.00	100.00
5 V / 2 A power supply	1	DIY	DYI
FC-51 IR sensor module	1	78.00	78.00
2N2222 transistor	4	10.00	40.00
1N4148 diode	2	5.00	10.00
1000 µF 16 V electrolytic capacitor	2	8.00	16.00
104 ceramic capacitor	5	5.00	25.00

Item	Qty	Unit Price (PHP)	Amount (PHP)
LTP-587G 16-segment display	1	48.00	48.00
NE556 timer IC	1	28.00	28.00
SG90 servo motor	1	168.00	168.00
10 kΩ trimmer potentiometer	2	1.00	2.00
Resistors (10 kΩ, 1 kΩ, 330 Ω, 100 kΩ), 19 pcs	19	1.00	19.00
Jumper and hook-up wires	5m	7.00	35.00
Model materials (board, paper, tape)	1	20.00	20.00
TOTAL			**Php589.00**

The estimated total cost of the project is PHP 589.00.

10. PHOTO DOCUMENTATION

